Math 54 — Week 1

Systems of linear equations; row reduction and echelon forms (Lay 1.1–1.2)

§1 This week

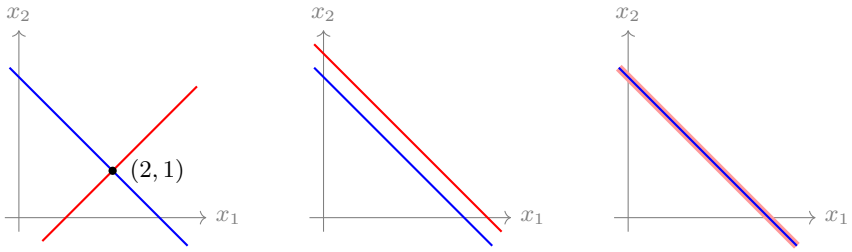
§1.1 Systems of linear equations (Lay 1.1)

The question. Three pairs of equations: $x_1 + x_2 = 3$ with $x_1 - x_2 = 1$; $x_1 + x_2 = 3$ with $2x_1 + 2x_2 = 7$; $x_1 + x_2 = 3$ with $2x_1 + 2x_2 = 6$. How many solutions does each pair have, and what do the solution sets look like?

Definition. A **linear equation** in the variables $x_1, \dots, x_n$ is an equation $a_1x_1 + \cdots + a_nx_n = b$ with numbers $a_1, \dots, a_n, b$. A **system of linear equations** (linear system) is a collection of such equations in the same variables. A **solution** is a list $(s_1, \dots, s_n)$ that makes every equation true when substituted. The **solution set** is the set of all solutions. Two systems are **equivalent** if they have the same solution set.

- $4x_1 - 5x_2 + 2 = x_1$ is linear (rearrange to $3x_1 - 5x_2 = -2$). $x_2 = 2\sqrt{x_1} - 6$ and $x_1x_2 + x_3 = 4$ are not: a variable appears under a root or multiplied by another variable.
- A system is **consistent** if it has at least one solution, **inconsistent** if it has none.

Two equations in two unknowns. Each equation $a_1x_1 + a_2x_2 = b$ with $(a_1, a_2) \neq (0, 0)$ is a line in the plane, and a solution of the system is a point on both lines.



$x_1 + x_2 = 3$, $x_1 - x_2 = 1$: exactly one solution

$x_1 + x_2 = 3$, $2x_1 + 2x_2 = 7$: no solution

$x_1 + x_2 = 3$, $2x_1 + 2x_2 = 6$: infinitely many

The three systems of the question. Two lines in the plane meet in one point, in no point (parallel lines), or in every point of a common line.

The question answered. The first pair has exactly one solution, $(2, 1)$: the lines have different directions. In the second pair, the second equation says $x_1 + x_2 = 3.5$, a parallel line, so there is no solution. In the third pair, the second equation is twice the first, so both describe the same line and every point on it is a solution.

Matrix notation. For the system

$$\begin{aligned} x_1 - 2x_2 + x_3 &= 0 \\ 2x_2 - 8x_3 &= 8 \\ 5x_1 - 5x_3 &= 10 \end{aligned}$$

coefficient matrix $\begin{bmatrix} 1 & -2 & 1 \\ 0 & 2 & -8 \\ 5 & 0 & -5 \end{bmatrix}$, augmented matrix $\left[\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 5 & 0 & -5 & 10 \end{array}\right]$.

The **coefficient matrix** holds the coefficients of the variables; the **augmented matrix** adds the right-hand sides as a last column. An $m \times n$ matrix has m rows and n columns. A missing variable has coefficient 0.

Shortcuts

- **Is an equation linear?** Each variable appears only to the first power, multiplied by a number, and never multiplied by another variable or inside a function. Reason: the definition. Instance: $3x_1 - 5x_2 = -2$ is linear; $x_1x_2 + x_3 = 4$ is not.
- **Two equations in two unknowns**: if the coefficient pairs (a_1, a_2) of the two equations are not proportional, there is exactly one solution; if they are proportional, there is none or infinitely many, depending on whether the right-hand sides are in the same proportion. Reason: proportional coefficient pairs describe parallel (or equal) lines. Instance: the question, where $(1, 1)$ and $(1, -1)$ are not proportional but $(1, 1)$ and $(2, 2)$ are.
- **Check a proposed solution** by substituting it into every equation. Instance: $(2, 1)$ gives $2 + 1 = 3$ and $2 - 1 = 1$.

Traps

- A solution must satisfy every equation: $(3, 0)$ satisfies $x_1 + x_2 = 3$ but not $x_1 - x_2 = 1$.
- A missing variable has coefficient 0, and its column must still appear in the matrix. The third equation $5x_1 - 5x_3 = 10$ above gives the row $(5, 0, -5 \mid 10)$, not $(5, -5 \mid 10)$.

§1.2 Elementary row operations (Lay 1.1)

The question. Elimination changes the equations. For $x_1 + x_2 = 2$, $x_1 - x_2 = 0$ (solution $(1, 1)$), which changes are safe: adding the first equation to the second? Multiplying the first equation by 0? Replacing the first equation by (first − second) and, at the same time, the second by (second − first)?

Definition. The **elementary row operations** on a matrix are:

(1) **Replacement**: replace a row by itself plus a multiple of another row.

(2) **Interchange**: swap two rows.

(3) **Scaling**: multiply all entries of a row by a *nonzero* constant.

Two matrices are **row equivalent** if one can be turned into the other by a sequence of elementary row operations.

Theorem (Row operations do not change the solution set (Lay 1.1))
If the augmented matrices of two linear systems are row equivalent, the two systems have the same solution set.

The idea

A row operation builds new equations from old ones, so anything that satisfied the old equations satisfies the new ones: no solution is lost. A solution could be gained only if the operation threw away information, and it does not: every row operation can be undone by another row operation, which also loses no solution.

Proof. 1. *Each operation keeps every solution.* Let $(s_1, \ldots, s_n)$ satisfy all the equations. Interchange and scaling by $c \neq 0$ produce the old equations in another order or multiplied by c, so they still hold. For a replacement, row i becomes (row i) $+ c\cdot$(row j). Substituting $(s_1, \ldots, s_n)$, the new left side is (left side of row i) $+ c\cdot$(left side of row j), which equals $b_i + cb_j$, the new right side.

2. *Each operation can be reversed by another row operation.* Undo a replacement by subtracting the same multiple, an interchange by swapping again, and scaling by c by scaling by $1/c$ (this is why $c = 0$ is not allowed).

3. If the new system had a solution that the old one lacked, applying the reversing operation would, by step 1, give a solution of the old system, which is that same list. So the solution sets are equal. For a sequence of operations, apply this to each step.

□

Worked example: solving by elimination.

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & 5 & 1 & 14 \\ -1 & 1 & 4 & 9 \end{array}\right] \xrightarrow[R_3+R_1]{R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & 8 \\ 0 & 3 & 3 & 12 \end{array}\right] \xrightarrow{R_3-3R_2} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & 8 \\ 0 & 0 & -6 & -12 \end{array}\right]$$

1. The aim is triangular form: x_1 only in the first equation, x_2 not below the second, and so on. Use the first row to clear the first column below it, then the second row to clear the second column below it.
2. The last row says $-6x_3 = -12$, so $x_3 = 2$. Back-substitute: $x_2 = 8 - 3x_3 = 2$, then $x_1 = 3 - 2x_2 + x_3 = 1$.
3. Check in the original equations: $1 + 4 - 2 = 3$; $2 + 10 + 2 = 14$; $-1 + 2 + 8 = 9$.

The question answered. Adding the first equation to the second is a replacement, so it is safe. Multiplying by 0 is not a row operation: it turns the first equation into $0 = 0$, and every point on the line $x_1 - x_2 = 0$ would then be a solution. The simultaneous change is not a sequence of row operations either: it gives $2x_2 = 2$ and $-2x_2 = -2$, which say only $x_2 = 1$, so $(5, 1)$ would become a solution.

Shortcuts

- **Several replacements in one step** are safe when they all use the same unchanged row, as in $R_2 - 2R_1$ and $R_3 + R_1$ above. Reason: doing them one after another gives the same result, because none of them changes R_1.
- **Avoid fractions early**: interchange to put a row with leading entry 1 on top, and scale only at the end. Instance: for
$$\left[\begin{array}{cc|c} 2 & 4 & 6 \\ 1 & 3 & 4 \end{array}\right],$$
interchange the rows first; then $R_2 - 2R_1$ needs no fractions.
- **Check** the final answer in the *original* equations, as in the worked example; an arithmetic slip shows up as a false equation.

Traps

- Scaling by 0 is not a row operation (it cannot be undone); the question shows that it adds solutions.
- Two operations in one step are safe only if neither uses a row the other changes. The simultaneous $R_1 \to R_1 - R_2$, $R_2 \to R_2 - R_1$ in the question adds solutions.

§1.3 Echelon forms (Lay 1.2)

The question. Two people row reduce

$$\begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}.$$

One uses $R_2 - 2R_1$; the other scales row 1 by 2 first and then uses $R_2 - R_1$. They get

$$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 2 & 4 \\ 0 & 2 \end{bmatrix}.$$

The results differ. What do they have in common, and is there one standard final shape from which the solutions can be read directly?

Definition. The **leading entry** of a row is its leftmost nonzero entry. A matrix is in **echelon form** if:

(1) all nonzero rows are above any rows of all zeros;

(2) each leading entry is in a column to the right of the leading entry of the row above it;

(3) all entries in a column below a leading entry are zero.

It is in **reduced echelon form** (RREF) if in addition:

(4) the leading entry in each nonzero row is 1;

(5) each leading 1 is the only nonzero entry in its column.

echelon: $\begin{bmatrix} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare \\ 0 & 0 & 0 & 0 \end{bmatrix}$ reduced echelon: $\begin{bmatrix} 1 & 0 & * & 0 \\ 0 & 1 & * & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ ($\blacksquare \neq 0$, $*$ any number)

Theorem (Uniqueness of the reduced echelon form (Lay 1.2, Thm. 1))
Each matrix is row equivalent to one and only one reduced echelon matrix.

The proof is in Lay, Appendix A (cited).

Definition. A **pivot position** of A is a position that holds a leading 1 in the RREF of A. A **pivot column** is a column of A containing a pivot position. A **pivot** is a nonzero number in a pivot position, used to create zeros.

The row reduction algorithm.

1. *Forward phase.* Take the leftmost nonzero column; it is a pivot column, with its pivot position at the top. (Row operations keep a zero column zero and cannot make a nonzero column zero, since they can be undone; so this column is also the first nonzero column of the RREF, whose leading 1 is in row 1.)
2. Choose a nonzero entry in that column as pivot, and interchange rows to move it to the top if needed.
3. Use replacement operations to create zeros in every position below the pivot.
4. Cover the row with the pivot (and all rows above it), and apply steps 1–3 to the rows that remain, until none are left. The matrix is now in echelon form.
5. *Backward phase.* Starting with the rightmost pivot and working up and to the left, create zeros above each pivot; scale each pivot row so the pivot is 1. The matrix is now in RREF.

Why the order works. In the forward phase, a covered row is never changed again, and the zeros below a pivot stay zero, because later replacements add multiples of rows that are already zero in that column. In the backward phase, the pivot row used to clear a column is zero in every pivot column to its right (those columns were cleared first), so clearing a column does not undo zeros already created.

Pivot positions are visible in every echelon form. An echelon form becomes the RREF by the backward phase (scalings and replacements that add multiples of lower rows), which never moves a leading entry. Since the RREF is unique, the leading entries of every echelon form of A are in the same positions: the pivot positions.

Worked example: interchange and backward phase.

$$\left[\begin{array}{ccc|c} 0 & 1 & 2 & 3 \\ 1 & 2 & 1 & 4 \\ 2 & 5 & 4 & 11 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 3 \\ 2 & 5 & 4 & 11 \end{array}\right] \xrightarrow{R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

1. Column 1 is the leftmost nonzero column, but its top entry is 0, so interchange to bring a nonzero entry up. The pivot positions are $(1,1)$ and $(2,2)$; in the original matrix the position $(1,1)$ holds 0, which is why an interchange was needed.
2. Backward phase: $R_1 - 2R_2$ gives the RREF

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

The question answered. Both results are echelon forms with leading entries in positions $(1,1)$ and $(2,2)$, and both reduce to the same RREF, the identity matrix.

Shortcuts

- **Pivot positions from any echelon form**: they are the leading entries; no need to reach the RREF. Reason: the paragraph above. Instance: the question.
- **Compare answers through the RREF**: two correct reductions of the same matrix reach the same RREF, even when the intermediate echelon forms differ. Instance: the question.

Traps

- An echelon form is not unique (the question shows two); the pivot positions and the RREF are.
- Each leading entry must be strictly to the right of the one above. This matrix is not in echelon form, because the entry 6 below the leading 4 must be cleared first:

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 6 & 7 \end{bmatrix}.$$

- A pivot position is a position in A, and A may hold 0 there before row reduction, as in the worked example.

§1.4 Solutions of linear systems (Lay 1.2)

The question. A system reduces to this augmented matrix. What are all its solutions? And how can we tell from an echelon form alone whether a system has no solution, one, or infinitely many?

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & -2 & 1 \\ 0 & 0 & 1 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Take an augmented matrix to reduced echelon form and read it back as equations.

Definition. The variables corresponding to pivot columns of the coefficient matrix are **basic variables**. The others are **free variables**.

Worked example: free variables.

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & -2 & 1 \\ 2 & 6 & 1 & -1 & 5 \\ -1 & -3 & 1 & 5 & 2 \end{array}\right] \xrightarrow[R_3+R_1]{R_2-2R_1} \left[\begin{array}{cccc|c} 1 & 3 & 0 & -2 & 1 \\ 0 & 0 & 1 & 3 & 3 \\ 0 & 0 & 1 & 3 & 3 \end{array}\right] \xrightarrow{R_3-R_2} \left[\begin{array}{cccc|c} 1 & 3 & 0 & -2 & 1 \\ 0 & 0 & 1 & 3 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

1. This is the matrix of the question, already in reduced echelon form. Pivot columns: 1 and 3. Basic variables x_1, x_3; free variables x_2, x_4.
2. Each nonzero row has exactly one basic variable, so solve each row for it: $x_1 = 1 - 3x_2 + 2x_4$ and $x_3 = 3 - 3x_4$.
3. **Parametric description of the solution set** (the **general solution**):

$$x_1 = 1 - 3x_2 + 2x_4, \qquad x_2 \text{ is free}, \qquad x_3 = 3 - 3x_4, \qquad x_4 \text{ is free}.$$

Every choice of x_2, x_4 gives a solution, and every solution arises this way. For example $x_2 = x_4 = 0$ gives $(1, 0, 3, 0)$.

Worked example: an inconsistent system.

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 2 & 5 & 4 & 2 \\ 3 & 7 & 7 & 4 \end{array}\right] \xrightarrow[R_3-3R_1]{R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -2 & 0 \\ 0 & 1 & -2 & 1 \end{array}\right] \xrightarrow{R_3-R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

The last row is the equation $0x_1 + 0x_2 + 0x_3 = 1$, which no numbers satisfy. The system is inconsistent; no further work is needed.

Theorem (Existence and uniqueness (Lay 1.2, Thm. 2))

A linear system is consistent if and only if the rightmost column of the augmented matrix is *not* a pivot column, that is, if and only if an echelon form of the augmented matrix has no row of the form $[\,0 \;\cdots\; 0 \;\; b\,]$ with $b \neq 0$. If the system is consistent, its solution set contains either exactly one solution (when there are no free variables) or infinitely many solutions (when there is at least one free variable).

The idea

Read the RREF as equations. A row $[\,0 \;\cdots\; 0 \;\; 1\,]$ is the false equation $0 = 1$. Every other nonzero row contains exactly one basic variable and can be solved for it in terms of the free variables, which nothing constrains.

Proof. 1. By the row operation theorem, the system has the same solutions as the system of its reduced echelon form, so we may read that form.

2. **Assume the rightmost column is a pivot column.** Then some row is $[\,0 \;\cdots\; 0 \;\; 1\,]$, the equation $0 = 1$. No solution.

3. **Assume the rightmost column is not a pivot column.** Then each nonzero row has its leading 1 in the column of a basic variable, and that basic variable appears in no other row (condition (5) of RREF). So the row says: basic variable = (a number) − (a combination of free variables). Choosing any values for the free variables and computing the basic variables this way satisfies every equation, so the system is consistent.

4. Every solution satisfies these row equations, so its basic variables are determined by its free variables. If there are no free variables, every variable has a single possible value: exactly one solution. If there is a free variable, different values of it give different solutions: infinitely many.

□

Using row reduction to solve a linear system. *Solving* a system means giving a parametric description of the solution set, or showing that it is empty.

1. Write the augmented matrix.
2. Row reduce to echelon form. If there is a row $[\,0 \;\cdots\; 0 \;\; b\,]$ with $b \neq 0$, stop: no solution.
3. Otherwise continue to reduced echelon form.
4. Write each nonzero row as an equation and solve it for its basic variable in terms of the free variables.
5. State the solution parametrically, naming the free variables as free.

The question answered. The worked example gives all the solutions. By the theorem, an echelon form decides the rest: no row $[\,0 \cdots 0 \;\; b\,]$ with $b \neq 0$ means consistent, and then free variables mean infinitely many solutions. The question's matrix has two free variables.

Shortcuts

- **Existence and uniqueness need only an echelon form**, not the RREF: look for a row $[\,0 \cdots 0 \;\; b\,]$ with $b \neq 0$, and count pivots among the variable columns. Reason: the pivot positions are the same in every echelon form. Instance: the inconsistent example stops at echelon form.

- **Symbolic echelon patterns** ($\blacksquare \neq 0$, $*$ any number) are read the same way. Instance: this augmented pattern has a pivot in its last column, so the system is inconsistent:

$$\left[\begin{array}{ccc|c} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & 0 & \blacksquare \end{array}\right].$$

- **Counting**: number of free variables = (number of variables) − (number of pivot columns of the coefficient matrix). Instance: the free-variable example has $4 - 2 = 2$ free variables.
- **More variables than equations**: each pivot lies in a different row, so there are at most as many pivots as rows. A consistent system with more unknowns than equations therefore has a free variable, hence infinitely many solutions. Condition: the system must be consistent. Instance: the free-variable example, with 3 equations in 4 unknowns.
- **Parameter questions** ("for which h is the system consistent?"): row reduce with the letters in place; the entries that end up in the last row decide. Instance:

$$\left[\begin{array}{cc|c} 1 & h & 2 \\ 4 & 10 & 5 \end{array}\right] \xrightarrow{R_2 - 4R_1} \left[\begin{array}{cc|c} 1 & h & 2 \\ 0 & 10-4h & -3 \end{array}\right].$$

  If $h = \frac{5}{2}$ this row is $0 = -3$: inconsistent. Otherwise $10 - 4h$ is a pivot and there is exactly one solution. Watch for a pivot that could be zero, and split into cases.

Traps

- A zero row $[0 \ \cdots \ 0 \ \ 0]$ does not by itself mean infinitely many solutions. The system $x_1 = 1$, $x_2 = 2$, $0 = 0$ has a zero row and exactly one solution. Count free variables.
- In the parametric description, each basic variable must be written in terms of free variables only. "$x_3 = 1 + x_2$" when x_2 is basic does not describe the solution set.
- Count pivots in the coefficient matrix, not the augmented matrix: a pivot in the last column means no solution, not an extra basic variable. The inconsistent example has 3 pivots in its augmented matrix but only 2 in its coefficient matrix.

§2 Homework this week

HW1, Sections 1.1–1.2, with a one-line hint for each exercise. (Sections 1.3–1.4 of HW1 are in the Week 2 handout.)

Section 1.1.

- **#1** (2×2 system). Eliminate x_1 from the second equation with a replacement, then back-substitute.
- **#8** (partly reduced augmented matrix). Scale the last pivot to 1, clear the entries above each pivot, then read off the solution set.
- **#9** (4×5 augmented matrix). Work upward from the bottom pivot (backward phase).
- **#13** (3×3 system). Reduce to triangular form as in the elimination example, then back-substitute.
- **#17** (check #13). Substitute into every *original* equation.
- **#21** (do three lines meet in a point?). Solve two of the equations, then test the third.
- **#25, #26** (values of h for consistency). One replacement, then the system is inconsistent exactly when a row $[0 \ 0 \ b]$ with $b \neq 0$ appears (as in P2).
- **#27** (row operations reversible?). For each of the three types, name the operation that undoes it (proof of the row operation theorem, step 2).
- **#32, #34** (inconsistent; equivalent). The definitions of inconsistent and of equivalent systems.

Section 1.2.

- **#1** (echelon or reduced echelon?). Check conditions (1)–(3), then (4)–(5).
- **#3** (pivot positions). Reduce to an echelon form; its leading entries mark the pivot positions, which you then circle in the original matrix.
- **#11** (general solution). Look at how the rows are related before reducing; then name the basic and free variables.
- **#13** (4×6 echelon matrix). Finish the backward phase, then solve each row for its basic variable.
- **#15 (or #17)** (check a general solution). Substitute the parametric solution, with the free variable left as a letter, into each original equation.
- **#19** (echelon patterns). Existence and uniqueness theorem: a pivot in the last column? then free variables?
- **#23** (choose h, k). Eliminate x_1, then split into cases on the coefficient of x_2 and the right side (as in P3).
- **#25** (two reduced echelon forms?). Lay 1.2, Theorem 1, together with the definition of row equivalence: can two different reduced echelon forms be row equivalent to each other?
- **#31, #32** ("solving"; free variables). What solving means (the method box "Using row reduction to solve a linear system"), and the theorem's two cases; remember the system may be inconsistent.

§3 Problem set

P1. Solve by row reduction and check your answer:

$$x_1 + x_2 + 2x_3 = 9, \quad 2x_1 + 4x_2 - 3x_3 = 1, \quad 3x_1 + 6x_2 - 5x_3 = 0.$$

P2. For which h is the system with this augmented matrix consistent?

$$\left[\begin{array}{cc|c} 1 & -3 & h \\ -2 & 6 & -5 \end{array}\right]$$

P3. For the system $x_1 + hx_2 = 1$, $3x_1 + 9x_2 = k$, find all h, k for which it has (a) no solution, (b) exactly one solution, (c) infinitely many solutions.

P4. Find the reduced echelon form and the general solution of the system with this augmented matrix:

$$\left[\begin{array}{ccccc|c} 1 & -2 & -1 & 3 & 0 & -2 \\ 0 & 0 & 1 & -4 & 2 & 5 \\ 2 & -4 & 0 & -2 & 2 & 1 \end{array}\right].$$

P5. True or false, with a reason. (a) A system of 3 equations in 4 unknowns can have exactly one solution. (b) If an echelon form of an augmented matrix has a row of zeros, the system has infinitely many solutions. (c) If the coefficient matrix of a system of 4 equations in 4 unknowns has a pivot in every column, the system has exactly one solution for every right-hand side.

P6. Find the polynomial $p(t) = a_0 + a_1 t + a_2 t^2$ whose graph passes through $(1, 4)$, $(2, 9)$, $(3, 18)$, by setting up and solving a linear system for a_0, a_1, a_2.

P7. Is the system $x + 2y = 5$, $3x - y = 1$, $2x + 3y = 8$, $x + y = 3$ consistent? If so, solve it.